Quiz #9

1. Determine whether each statement is **True or False**. Briefly explain your answer.
   (a) A test gives $P = 0.03$. This means there is a 3% probability that the null hypothesis is true.
   (b) Results with $P = 0.049$ and $P = 0.051$ can provide nearly the same evidence against H_0, even though only one is statistically significant at $\alpha = 0.05$.
   (c) If a result is not statistically significant, the population difference must be zero.
   (d) A very small difference can be statistically significant when the sample size is large, even if the difference is too small to matter in practice.

2. A laboratory tests 20 proposed process changes. Assume none has a real effect and each test has a 5% chance of a false positive.
   (a) How many statistically significant results would you expect, on average, if each test uses $\alpha = 0.05$?
   (b) The laboratory reports only the smallest P-value, $P = 0.02$, and declares that change effective. Why is this reasoning misleading?
   (c) Use a Bonferroni correction to limit the probability of at least one false positive across the 20 tests to at most 0.05. What cutoff should each test use? Does $P = 0.02$ meet it?

3. For each situation, assess the proposed analysis.
   (a) An instructor has accurate grades for *every* student in two specified course offerings. The percentages receiving an A were 30% and 35%. Is a sampling-based significance test needed to determine whether these two complete cohorts had different percentages? Explain.
   (b) A laboratory tests each of the same 10 sensors at three temperatures, producing 30 readings. It treats the temperatures as three independent groups and applies an ordinary one-way ANOVA. Which assumption is questionable, and why?

4. A laboratory randomly assigns 18 independently prepared specimens to three curing temperatures, with 6 specimens per temperature. It measures tensile strength (MPa). Assume the usual one-way ANOVA conditions hold.

   Python produces the following ANOVA table, with five entries replaced by blanks. Leave the two `NaN` entries unchanged.

```
                  df  sum_sq  mean_sq     F    PR(>F)
C(temperature)  ____    24.0     ____  ____  0.012175
Residual        ____    30.0     ____   NaN       NaN
```

(a) State H_0 and H_A for comparing the three population mean strengths.

(b) Fill in all five blanks in the table. Show how you calculate the mean squares and the F statistic.

(c) Use the P-value in the table to state your conclusion at $\alpha = 0.05$. Does it show that every pair of population means differs?

5. An engineer randomly samples 120 parts from a factory. A CSV file records each part's production line (A or B) and inspection result (Fail, Pass, or Rework). Assume observations are independent. Parts were not randomly assigned to production lines.

```
import pandas as pd
from scipy import stats

parts = pd.read_csv("part_inspections.csv")
observed = pd.crosstab(parts["line"], parts["inspection"])
result = stats.chi2_contingency(observed, correction=False)
print(observed)
print(f"chi2 = {result.statistic:.2f}, P = {result.pvalue:.4f}")
```

The code prints:

```
inspection  Fail  Pass  Rework
line
A              9    36      15
B             21    24      15
chi2 = 7.20, P = 0.0273
```

(a) What does `pd.crosstab()` calculate? State the null and alternative hypotheses tested by this code.

(b) Under H_0, what is the expected number of failed parts from Line A? What are the test's degrees of freedom?

(c) What is your conclusion at $\alpha = 0.05$? Does this result establish that production line *causes* inspection outcomes to differ?